

Palitizer / PLC_1 [CPU 1211C DC/DC/DC] / Program blocks

FB_Motor [FB6]

FB_Motor Properties							
General							
Name	FB_Motor	Number	6	Type	FB	Language	SCL
Numbering	Automatic						
Information							
Title		Author		Comment		Family	
Version	0.1	User-defined ID					

FB_Motor									
Name	Data type	Default value	Retain	Accessible from HMI/OPC UA/Web API	Writ-able from HMI/OPC UA/ Web API	Visible in HMI engi-neering	Setpoint	Supervi-sion	Comment
▼ Input									
▼ i_stState	"UDT_SysStatus"		Non-retain	True	True	True	False		
xModeAuto	Bool	false	Non-retain	True	True	True	False		
xModeManual	Bool	false	Non-retain	True	True	True	False		
xRunning	Bool	false	Non-retain	True	True	True	False		
xPaused	Bool	false	Non-retain	True	True	True	False		
xFaulted	Bool	false	Non-retain	True	True	True	False		
iStateID	Int	0	Non-retain	True	True	True	False		
iCurrentLayer	Int	0	Non-retain	True	True	True	False		
iTargetLayers	Int	0	Non-retain	True	True	True	False		
iTotalBoxes	DInt	0	Non-retain	True	True	True	False		
iTotalPallets	DInt	0	Non-retain	True	True	True	False		
wAlarmWord	Word	16#0	Non-retain	True	True	True	False		
i_xAutoCmd	Bool	false	Non-retain	True	True	True	False		
i_xManualCmd	Bool	false	Non-retain	True	True	True	False		
i_xInterlock	Bool	false	Non-retain	True	True	True	False		
i_xThermalFault	Bool	false	Non-retain	True	True	True	False		
▼ Output									
q_xRunFwd	Bool	false	Non-retain	True	True	True	False		
InOut									
Static									
Temp									
Constant									

```
0001 // Check motor hardware health first (assuming NC thermal contact)
0002 IF #i_xThermalFault THEN
0003
0004     // check mechanical interlock
0005     IF #i_xInterlock THEN
0006
0007         // auto mode
0008         IF #i_stState.xModeAuto AND #i_stState.xRunning THEN
0009             #q_xRunFwd := #i_xAutoCmd;
0010
0011         // manual mode
0012         ELSIF #i_stState.xModeManual THEN
0013             #q_xRunFwd := #i_xManualCmd;
0014
0015         // off / faulted
0016         ELSE
0017             #q_xRunFwd := FALSE;
0018         END_IF;
0019     ELSE
0020         // Interlock broken
0021         #q_xRunFwd := FALSE;
0022     END_IF;
0023 ELSE
0024     // Motor thermal overload tripped!
0025     #q_xRunFwd := FALSE;
0026 END_IF;
```

Symbol	Address	Type	Comment
#i_stState.xModeAuto		Bool	
#i_stState.xModeManual		Bool	
#i_stState.xRunning		Bool	
#i_xAutoCmd		Bool	
#i_xInterlock		Bool	
#i_xManualCmd		Bool	
#i_xThermalFault		Bool	
#q_xRunFwd		Bool	